

A Multi-Kilowatt Low-Profile GaN Inverter for Light Electric Vehicles and High-Power Tools

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Abstract—An ultra-compact three-phase inverter with 8 devices per switch is presented for a combined output power of 7.3 kW at 48 V. Due to the use of Gallium-Nitride (GaN) transistors the design can operate at a frequency of 50 kHz and above without significant loss penalty. Parallel switching of the GaN device is enabled by symmetrical a layout with a “butterfly” structure. The high switching frequency enables to shrink the size of the DC-link capacitor to an overall size of 72 by 132 mm with a height of as low as 4 mm depending on the chosen cooling solution resulting in an overall volume of 0.1 liter. The measured peak efficiency of over 99.3% enables small heatsink size and high output power with minimal airflow. Under free convection and device temperature limited to 100°C the inverter reached an output power of 4kW with efficiency above 99% across the operation range. Using active cooling the capable of delivering 7.3kW while keeping device temperatures around 60°C.

I. INTRODUCTION

An increasing amount of applications that used to run off small off-road engines (SORE) are now transitioning to electric due to environmental concerns as well as legislative initiatives. Typical applications include larger power and gardening tools as well as small and light electric vehicles (LEVs). For these applications there is a strong need for highly efficient, small and light-weight motor drives to alleviate the increased weight introduced by the batteries. Power requirement on these motor drives is in the multi-kilowatt range, i.e. from 2 to 19kW with a battery voltage ranging typically from 48 to 120 V. Conventional inverter designs based on silicon (Si) MOSFET technology struggle to meet expectation in terms of size and weight [1]. Gallium-Nitride (GaN) High Electron Mobility Transistors (HEMTs) are a potentially attractive solution to meet the increased demand for performance [2]. However, currently these devices are not widely available in large die sizes which makes it difficult to address high current application with a single device. To overcome this issue the literature reports paralleling of up to four GaN HEMTs successfully [3], [4], [5], [6]. In addition, several concepts have been proposed to increase the system output power by paralleling motor drives[7], using poly-phase [8], [9], [10] or multi-level systems [11], [12]. However, these solutions come with a lot of added complexity which is not always desirable. This work presents a high-current three-phase motor drive with 8 GaN HEMTs per switch in parallel. In particular, the design is capable of maintaining high switching-speed and low switching-loss thanks to a consequential low-

impedance design. To the authors knowledge this is the first comprehensive study on paralleling more than four devices effectively.

II. FUNDAMENTAL CONSIDERATIONS FOR PARALLELING ON DEVICE-LEVEL

Similar to Si MOSFETs, GaN HEMTs are majority carrier devices and exhibit a positive temperature coefficient for $R_{DS(on)}$ with respect to the junction temperature T_j . For a 100 V Si MOSFET and a comparable GaN HEMT the $R_{DS(on)}$ vs. T_j is shown in **Fig. 1**. The GaN HEMT clearly exhibits an increase temperature coefficient over the Si MOSFET. For paralleling this behavior is actually desirable as it helps with steady-state current balancing. In case of a current imbalance, the device carrying the most current will heat-up more which leads to higher resistance and in turn to lower current through that device.

In addition to the static $R_{DS(on)}$, the transfer characteristic as well as transconductance $\Delta I_D / \Delta V_{GS}$ are important characteristics for dynamic current sharing. **Fig. 2** compares the behavior for aforementioned devices. For comparison the plot has been normalized the threshold voltage at 25°C of 3.3 and 2.1 V for the MOSFET and HEMT respectively. At 25°C the GaN HEMT shows a much higher transconductance than the MOSFET. However, the transconductance is also more strongly affected by temperature, effectively slowing down the switching transient. This temperature coefficient helps to balance dynamic current sharing, as hotter devices will naturally carry less of the hard-switching burden than colder devices.

Furthermore, **Fig. 2** provides an indication of the sensitivity of V_{th} towards temperature for the MOSFET and HEMT. While I_D remains negligible regardless of temperature for the HEMT, the MOSFETs drain current increases notably with temperature when V_{th} is left at 3.3V. In other words, to keep the MOSFET at the threshold of conduction, V_{GS} needs to be decreased considerably with temperature. The temperature coefficient $\Delta V_{th} / \Delta T_j$ is -3.75 and only -1.1 mV/K for the considered MOSFET and HEMT respectively. Further information can be found in a previous publication [13].

allows the driver to output a peak current of 4 A for sourcing and sinking during switching. Furthermore, an active Miller clamp boost the sinking output to 7 A once the driver detects that the gate voltage has fallen below 0.4 V. This feature avoids spurious switching due to the Miller effect. Added robustness against induced turn-on is provided by an integrated charge pump within the driver, which internally generates an adjustable negative rail to ensure that all GaN HEMTs remain off at the proper times [14]. It is important to consider that GaN HEMTs have a reverse conduction mechanism that mimics a MOSFET body diode, with the exception that the "body diode" voltage drop is directly proportional to the negative voltage applied. Therefore, applying a more negative off-state voltage than necessary produces higher losses during the dead-time. Furthermore, when the high-side driver is supplied via bootstrapping as in **Fig. 3** this mechanism increases the effective voltage being applied during deadtime. A Zener diode (D_{ZNR}) is therefore included in parallel to C_{VDD} of the high-side. The amount of over-charging will depend on the (parasitic) impedance and deadtime.

The high-side gate traces are shown in **Fig. 4**. Great care was given to the symmetrical design of the gate loops via the "butterfly" structure with the gate driver in the center. Implementation of the low-side traces is similar. Moreover, this layer also contains part of the switch-node for which the output current is indicated. A second switch-node layer is placed directly on top providing a tightly coupled, low-inductance return loop for the gate current as well as lower resistance for the output current.

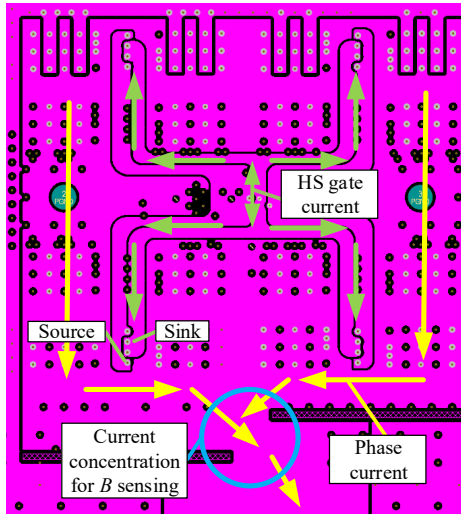


Fig. 4. PCB layer 5 which includes the switch node, the gate path (source and sink) and concentration for phase current to increase the $\vec{B}$ -field for current sensing.

Fig. 5 shows the simulated impedance of the 8 different gate paths. It can be clearly seen that all loops match very well along the considered frequency range. At 500 MHz the inductance of the 8 loops is about 1 nH with less than 17% difference in impedance.

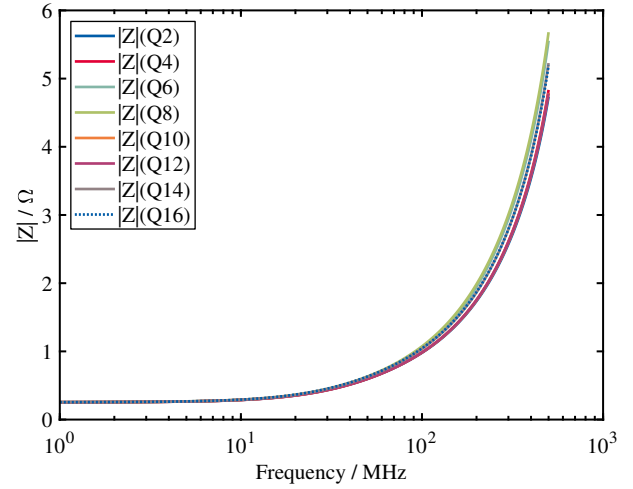


Fig. 5. Impedance of the gate loop for all low-side device with respect to the central gate driver.

D. Sizing of the DC-Link

As [15] has shown, the RMS current of the DC-link in a voltage-source-inverter is primarily dependent on the output current of the inverter and power factor. As first approximation for sizing the DC-link, the capacitor RMS current I_{Cin} can be derived from the phase current I_{phs} as

$$I_{Cin} = 0.65 \cdot I_{phs}. \quad (1)$$

For a targeted 130 A_{rms} this results in about 85 A_{rms} current going through the DC-link. For a single capacitor this would be considerable. However, thanks to the low switching-loss of GaN HEMTs, frequency can be increased to a point where MLCCs become viable instead of aluminum electrolyte capacitors. With respect to their size the current handling capability of MLCCs is much higher and paralleling them is straightforward. Paralleling many MLCCs has the added benefit of reducing the parasitic inductance, thus reducing ringing and improving current sharing [13].

In addition to the current handling capability, the voltage ripple needs to be considered. The amount of total DC-link capacitance C_{DC} depends on the acceptable voltage ripple $\Delta v_{pp,max}$, the peak phase output current $\hat{I}_{phs}$ and the switching frequency of the inverter f_{sw} . It can be estimated as follows

$$C_{DC} \geq \frac{\hat{I}_{phs}}{4f_{sw}\Delta v_{pp,max}}. \quad (2)$$

Given a switching frequency f_{sw} of 50 kHz and an acceptable voltage ripple of 5% at 48 V (2.4 V_{pp}), C_{DC} needs to be at least 375 μ F. Of course this can be further refined depending on requirements for size, output power and efficiency.

E. Thermal Considerations

To enable high power motor drives, an effective thermal solution is required. For this reason all GaN devices have been placed on one side of the PCB to facilitate easy

attachment to a heatsink. Crucially, all components have been chosen not to exceed the height of the HEMTs. The dual-sided cooling package of the GaN HEMTs provides an effective cooling path upward to the heatsink as well as downward to the PCB. Off-the-shelf solutions like push-pin heatsinks or custom cooling solutions can be implemented easily where the entire top-side of the PCB is attached to a cold-plate. Latter solutions can also take advantage of additional thermal paths through the MLCC capacitors for even greater effect. Several mounting holes in the PCB design support this approach.

IV. IMPLEMENTED PROTOTYPE

The prototype is implemented on a six-layer PCB with 70 μm of copper on each layer and a total size of 132 by 72 mm. Each half-bridge consist of 8 Infineon CoolGaN™ 100 V 3 m Ω HEMTs in parallel for the low-side switch as well 8 devices in parallel for the high-side switch. Each switch is driven by an Infineon EiceDRIVER™ 1EDN7116G. Current sensing is done through in-phase sensing based on the Infineon XENSIV™ TLE4972 magnetic current sensor. The inverter is controlled by an Infineon XMC™ 4400 industrial microcontroller with 150 ps high-resolution PWM for precise control.

Fig. 6 shows a close-up image of a single half-bridge. The active components (HEMTs, gate driver etc.) occupy an area of 30 by 25 mm and the total size of the half-bridge incl. DC-link is 47 by 37 mm. A detailed description of the layer stack-up can be found in [13].

The total DC-link of 580 μF is split up across the three phases. Each phase contains 87 MLCCs in 1206 with 2.2 μF each plus an additional 42 MLCCs in 0603 with 100 nF each for the high frequency switching current and are located at the top and bottom of the half-bridge. The nominal capacitance of the DC-link is 580 μF but care must be taken to account for DC-bias. Therefore, additional footprints for leaded capacitors are also provisioned.

Components on either side of the PCB are 1 mm or less resulting in an overall thickness of less than 4 mm. Several mounting holes can be seen which allow either off-the-shelf heatsinks to be mounted or the entire PCB can be attached to a cold plate. This can also provide additional paths for heat to leave the PCB via the MLCCs.

The negative V_{GS} of the gate driver was limited to -1.5 V providing generous margin against parasitic turn-on. In paralleled designs it is often common-practice to place individual gate resistors in front of each gate to dampen oscillations caused by the Miller charge. For this design R_{G} was specified at 1 Ω with no observable degradation in switching speed. The optional additional C_{GS} was chosen to be 270 pF. Further refinements to the negative V_{GS} , R_{G} and C_{GS} are possible depending on application requirements.

A 5.1 V Zener diode was used on the high-side to limit overcharging. To further limit the gate current R_{drv} was chosen to be 1 Ω .

Fig. 7 shows the full PCB with three push-pin heatsinks with 30 by 30 mm size and a fin height of 10 mm. Because the top-

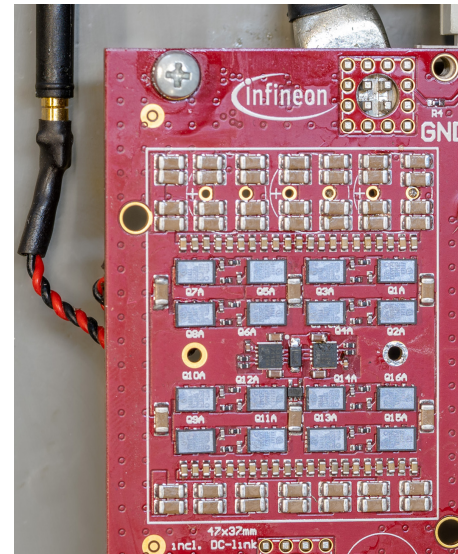


Fig. 6. Close-up image of a single half-bridge.

side thermal pad of the GaN HEMT is internally connected to its source terminal, an electrically isolated thermal interface material (TIM) is required. In this case a ultra-soft TIM with an uncompressed thickness of 500 μm is used. Compression of the TIM is limited to around 20% by the limited spring force. When attaching the entire top-side of the PCB to a custom heatsink with screws, higher compression and lower thermal resistances are possible.

Even though a heatsink enhances the thermal dissipation capability of the inverter it is not always needed. **Sec. V-B** discusses the power capability with and without heatsink and airflow.

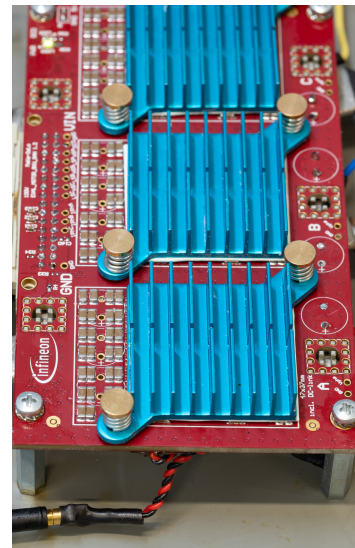


Fig. 7. Implemented prototype with spring loaded heatsink applied

V. EXPERIMENTAL VERIFICATION

Verifying correct operation of 8 paralleled devices in a three-phase system is not trivial. To facilitate experimental evaluation the converter, it is first operated as a simple buck converter with 50% duty cycle. This makes efficiency measurements straightforward and the switching behavior for individual current levels can be analyzed more easily.

In order to verify correct operation of all 8 devices in parallel, thermal imaging is used. In a previous work current sharing between the individual half-bridges was analyzed and only showed minor variations [13].

Probing and waveform acquisition was done with a 1 GHz passive probe and scope with equal bandwidth and 6.25 GS/s. For ease of use during initial testing, probes with MMCX tips were used. These allow to quickly connect to different signals. The PCB exposes important signals on dedicated test pads to which short, twisted wires were soldered onto, see left side of **Fig. 6** and bottom side of **Fig. 7**. High-side V_{GS} measurements use the same connection scheme with an optically isolated probe with 800 MHz bandwidth. Even though this setup is highly convenient, it was found that some noise can still couple into the twisted wires. Especially for small signals such as V_{GS} and in particular for the floating high-side V_{GS} this must be considered.

A. Operation As a Buck Converter

To confirm safe voltage levels of the high-side driver, V_{DD} was measured with an isolated multimeter at 50 kHz, 25 ns deadtime and a negative V_{GS} of -1.5 V. **Fig. 8** shows V_{DD} in relation to the output current up to 130 A at 48 V. With increasing output current the “diode-like” operation of the GaN device gets more and more pronounced which leads to an increasing voltage being applied to the high-side driver during deadtime. However, as long as V_{GS} stays within acceptable limits this behavior is beneficial by providing more symmetrical supply voltages between high and low-side. Under the considered conditions the high-side V_{DD} reaches about 4.77 V at 130 A which almost fully compensates for the bootstrap diode drop while staying within recommend voltage levels. Higher deadtime value lead quickly to over-charging the high-side V_{DD} and excessive power loss in the Zener diode (not shown).

Fig. 9 and **Fig. 10** show the falling and rising edge of the switch-node voltage V_{sw} respectively. Due to the operation as a sync. buck, all falling edges are either partial or full ZVS transitions while rising edges are (partially) hard switching. The switch-node probing point was attached to the middle of the switch-node, thus giving an indication of the overall switching behavior instead of one device specifically. V_{GS} was measured for the high-side device Q9 and the low-side device Q10 as close as possible on test points directly beneath the devices (c.f. **Fig. 6** for designators and device placement).

Fig. 9 shows a set of waveforms ranging from 0 to 130 A output current with little ringing on all waveforms, due to the soft ZVS transition. At 130 A the slew rate reaches 13 V/ns.

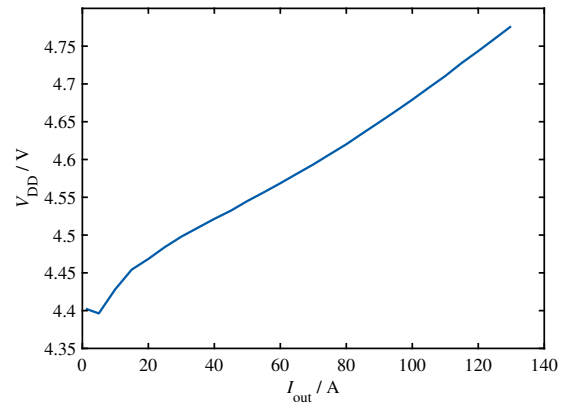


Fig. 8. High-side driver supply voltage V_{DD} vs. output current in buck configuration.

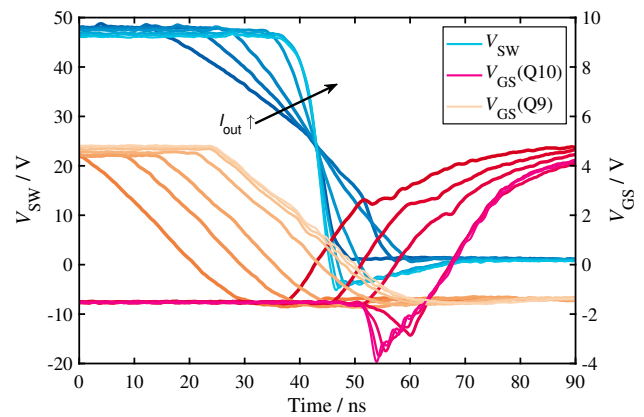


Fig. 9. Falling switch-node voltage V_{sw} as well as low and high-side V_{GS} of Q9 and Q10 respectively. The trend of waveforms from 0 to 130 A is indicated by increasing brightness.

Fig. 10 shows the hard turn-on of the high-side device. Before the voltage rises the diode-like reverse conduction of the HEMT can be seen and how it increases with output current. Compared to **Fig. 9** more ringing can be observed in all three waveforms. However, as stated previously this is in part to noise being coupled into the probing setup, especially on the high-side V_{GS} measurement.

With increasing output current the switch node voltage shows a small dip between 30 and 40 V. This behavior has been observed in the literature [16] for other GaN devices as well, due to the nonlinearity of the Miller effect. At 130 A a peak slew rate of 25 V/ns is reached.

The efficiency of the half-bridge running in buck mode was evaluated by measuring the input voltage and current V_{in} and I_{in} as well as the output current I_{out} and the average output voltage V_{out} on the screw terminal of the PCB. This way external losses associated with external wiring or inductors are not captured in the efficiency.

Experiments were carried out for frequencies from 50 to 125 kHz in 25 kHz increments up to a temperature of 100°C

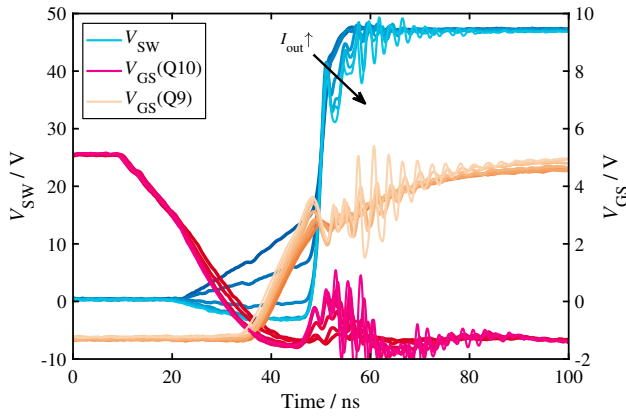


Fig. 10. Rising switch-node voltage V_{SW} as well as low and high-side V_{GS} of Q9 and Q10 respectively. The trend of waveforms from 0 to 130 A is indicated by increased brightness.

on the on-board temperature sensor. Slight variations in the achieved temperature are due to the discrete steps in output current. All experiments were carried out without heatsink but airflow of 4 m/s which allows to observe the HEMTs exposed substrate with a thermal camera. Additional bulk capacitance was used to compensate for the lower effective frequency present at the DC-link compared to when all three half-bridges would have been used.

One of the outside half-bridge was investigated instead of the center one. This presents a worse case scenario with asymmetrical heat flux and higher thermal gradient.

As can be seen from **Fig. 11**, peak efficiency is well above 99% for 50 kHz switching frequency (measured 99.3%). Increasing the frequency reduces efficiency slightly until at 125 kHz only 99.1% are reached. A similar picture can be seen for the maximal output current. At 50 kHz an output current of 130 A can be reached. Higher frequencies successively reduce the output current capability to 125, 120 and 115 A respectively.

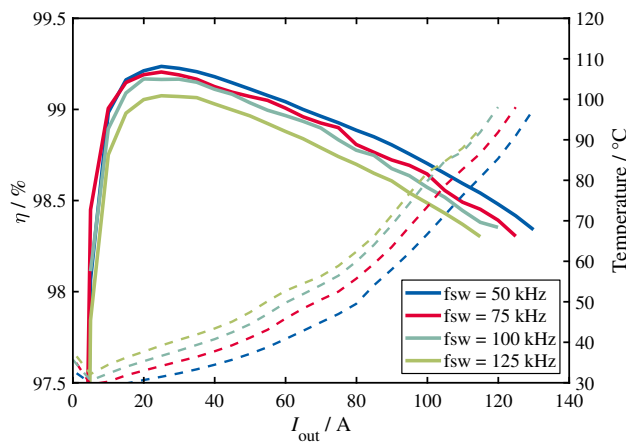


Fig. 11. Efficiency of a single phase running as buck converter with 50% duty cycle at 48 V.

Additionally to the half-bridge efficiency, thermal images were captured. For an output current of 125 A the thermal image is shown in **Fig. 12**. Due to physical location on the edge of the board a thermal gradient can be observed with the right side being cooler than the left due to a large, non-heated surface area of the PCB. Furthermore, a local hotspot can be seen, due to the current concentration structure for the $\vec{B}$ -field sensing. Apart from these thermal asymmetries, device temperature is consistent and in-line with expectation. No outlier were observed. The hottest devices being Q12 with 102°C and the coldest device Q1 with 65°C. Despite this notable thermal gradient correct operation of the half-bridge could be demonstrated. In a three-phase inverter application and esp. with a heatsink thermal gradient will be significantly smaller.

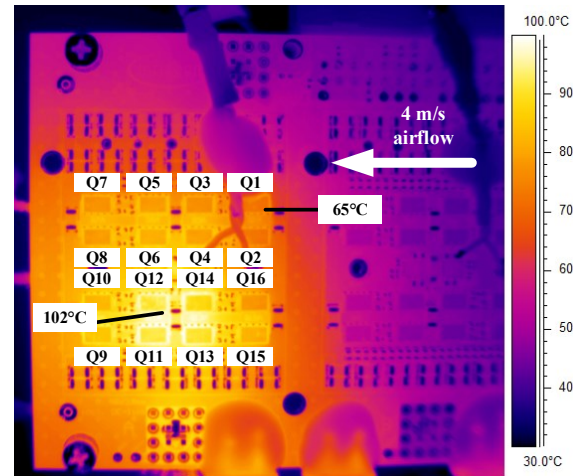


Fig. 12. Thermal image of the converter running at 48 V and 50 kHz with 125 A output current in single-phase buck-mode.

B. Operation as a Three Phase Motor Drive

After correct operation of a single half-bridge has been confirmed this section covers operation as a three phase inverter.

For accurate efficiency measurement, a passive inductive-restive load was chosen to emulate a full motor system. **Fig. 13** shows the inverter efficiency versus RMS output current for different cooling solutions when operated at 48 V with 50 kHz switching frequency and 25 ns deadtime. Peak temperature was limited to 100°C in all experiments based on the on-board sensor.

In free air with natural convection an output current per phase of 67 A_{rms} was achieved with measured inverter loss of 25 W. At maximum modulation index with space vector modulation this represents a total output power of 4 kW a restive load. In the final application output power can be slightly less, depending on the amount of reactive power (i.e. $\cos(\phi)$) the inverter has to support at the motor side. Output current can be increased to 73 A_{rms} or 4.3 kW when using the small, off-the-shelve push-pin heatsinks shown in **Fig. 7**. Due

to tight fin spacing and short height only moderate increase in heat dissipation to 30 W was achieved.

Significantly more output current can be achieved when forced convection is used. With a moderate airflow of 4 m/s without heatsinks the output current capability per phase reaches 102 A_{rms} or 6 kW. Under these conditions the board is able to dissipate 52 W. Current can be further increased with the aforementioned heatsink to 120 A_{rms} or 6.6 kW at which the inverter dissipates 70 W. Lastly, considerable airflow of 10 m/s was applied and the experiment was run to the setups capability of 125 A. Under these conditions the temperature sensor reported around 60°C at peak output current. The lower temperatures results in a notable increase in efficiency. This is caused primarily caused by lower $R_{DS(on)}$ as well as slightly faster slewrates, due to higher transconductance.

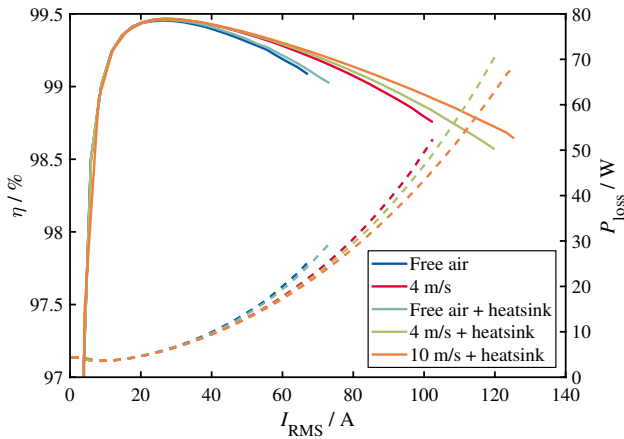


Fig. 13. Efficiency of the inverter running as a three phase system at 48 V with 50 kHz switching frequency.

Fig. 14 shows the thermal performance of the inverter at 102 A_{rms} output current with an airflow of 4 m/s. For each half-bridge the maximum and minimum temperatures are reported. Naturally, the side facing the air flow is the coldest with a peak temperature of 78°C, the other two half-bridges show a very similar thermal profile with peak temperatures of 94°C each. The thermal gradient is within expectations and confirms correct operation of the paralleled half-bridges. No significant anomalies that would indicate imbalance in power could be detected.

VI. CONCLUSION

This work presents an ultra-compact three-phase inverter with eight 100 V CoolGaN™ Schottky Gate HEMTs in parallel per switch position. Due to the temperature coefficients of the GaN HEMT, paralleling a large number of devices is fundamentally no more difficult than with Si MOSFETs. Balanced gate and power-loops are ensured by the presented highly symmetrical “butterfly” layout. Furthermore, design considerations for the gate driver and DC-link are presented to fully utilize the benefits of the HEMTs in a motor drive application.

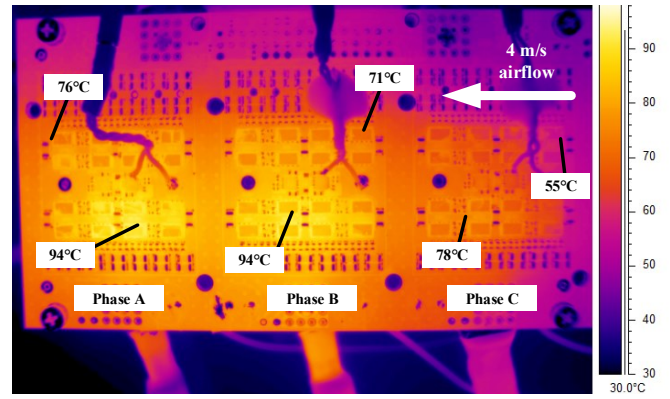


Fig. 14. Thermal image of the inverter running at 50 kHz and 100 A_{rms} output current with 4 m/s of airflow. Minimum and maximum temperature of each half-bridge is indicated.

The implemented PCB has a total size of 132 by 72 mm with a maximum thickness of 4 mm. Using natural convection and no heatsink, this configuration achieves 4 kW of output power at 48 V while keeping efficiency above 99%. With moderate airflow of 4 m/s the output power can be increased to 6 kW. Adding a heatsink with 10 mm height enabled the output power to reach 6.6 kW with the same 4 m/s forced air, and up to 7.3 kW with 10 m/s air flow. Preliminary testing indicates much higher power levels are feasible, especially when the PCB is mounted to a coldplate via screws for increased compression of the TIM and lower thermal resistance. All device temperatures stayed within expectation, no abnormal temperature distribution could be detected.

This work has shown how a GaN-based motor drive offers very high efficiency across a wide power range, making it clearly very attractive for electrification of applications that currently use a conventional combustion engine. Lower weight and smaller size add further value to applications like handheld high-power tools, as well as increasing the range of light electric vehicles.

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